

| Question | Answer                                                                                                                                                                    | Marks       |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3(a)(i)  | $\sigma = 0.72 \times 10^9$                                                                                                                                               | <b>C1</b>   |
|          | $\text{force} = \sigma \times A$<br>$= 0.72 \times 10^9 \times \pi \times (1.2 \times 10^{-3} / 2)^2$                                                                     | <b>C1</b>   |
|          | $= 810 \text{ N}$                                                                                                                                                         | <b>A1</b>   |
|          | <b>or</b><br><br>Young modulus = gradient of graph<br>e.g. $= 0.80 \times 10^9 / 6.0 \times 10^{-3}$<br>$= 1.33 \times 10^{11}$                                           | <b>(C1)</b> |
|          | $\text{force} = \text{Young modulus} \times \text{strain} \times A$<br><br>$= 1.33 \times 10^{11} \times 5.4 \times 10^{-3} \times \pi \times (1.2 \times 10^{-3} / 2)^2$ | <b>(C1)</b> |
|          | $= 810 \text{ N}$                                                                                                                                                         | <b>(A1)</b> |

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|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 3(a)(ii) | $E_{(P)} = \frac{1}{2} Fx$<br>or $E_{(P)} = \frac{1}{2} kx^2$ <b>and</b> $F = kx$                                                                                          | <b>C1</b>   |
|          | $x = 2E_P / F$<br>$x = 2 \times 0.31 / 810$<br>$x = 7.7 \times 10^{-4}$                                                                                                    | <b>C1</b>   |
|          | $L = x / \epsilon$<br>$L = 7.7 \times 10^{-4} / 5.4 \times 10^{-3}$<br>$L = 0.14 \text{ m}$                                                                                | <b>A1</b>   |
|          | <b>or</b><br>$E_{(P)} = \frac{1}{2} Fx$<br>or $E_{(P)} = \frac{1}{2} kx^2$ <b>and</b> $k = EA/L$                                                                           | <b>(C1)</b> |
|          | $x = 2E_P / EA \epsilon$<br>$x = 2 \times 0.31 / (1.33 \times 10^{11} \times \pi \times (1.2 \times 10^{-3} / 2)^2 \times 5.4 \times 10^{-3})$<br>$x = 7.6 \times 10^{-4}$ | <b>(C1)</b> |
|          | $L = x / \epsilon$<br>$L = 7.6 \times 10^{-4} / 5.4 \times 10^{-3}$<br>$L = 0.14 \text{ m}$                                                                                | <b>(A1)</b> |
| 3(b)     | A straight line, passing through the origin with a larger gradient than wire X.                                                                                            | <b>M1</b>   |
|          | Gradient of the line is twice the gradient of wire X.                                                                                                                      | <b>A1</b>   |

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